WORKED SOLUTION

STEP I 2008, Question 13

Probability and Statistics • Circular Arrangements

STEP I 2008, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q13

The key idea

At a round table only the people's positions relative to each other matter, so we fix one person's chair and arrange the other five around them. That gives $5! = 120$ equally likely arrangements, and every probability is a count out of 120. A couple who must sit together can be glued into a single "block" that has two internal orders. For "exactly" and "none" we either count directly or use inclusion–exclusion over the events "couple k sits together".

Solution

Call the couples A, a; B, b; C, c. Rotating everyone one place round the table gives the same seating relative to each other. So we fix A's chair and seat the other five people in the remaining chairs in order going clockwise. This gives $5! = 120$ arrangements, all equally likely.

If we labelled the chairs instead, there would be $6! = 720$ arrangements. Each relative arrangement then appears 6 times, once for each chair A could occupy, so the probabilities are the same.

A useful general count: n objects round a circle can be arranged in $(n-1)!$ ways (fix one object, then order the rest).

Part (i)

If every couple sits together, glue each couple into a block. Then we have 3 blocks round the table, which can be arranged in $(3-1)! = 2$ ways. Inside each block the husband can sit on either side of his wife, which gives $2^3 = 8$ choices. So

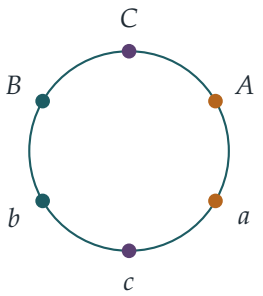
$$\mathrm{P}(\text{all three couples together}) = \frac{2 \times 8}{120} = \frac{16}{120} = \frac{2}{15},$$

as required.

Part (ii)

First choose which couple is *not* together. There are 3 choices. Say it is C, c, so A, a and B, b are together but C and c are not.

Glue Aa and Bb into blocks. Round the table we now have 4 units, Aa, Bb, C, c, which can be arranged in $3! = 6$ ways. Fix C and look at where c goes among the other three places of this 4-unit circle. In two of them c is next to C, and in one c is directly opposite. So C and c are apart in exactly $6/3 = 2$ of the 6 arrangements. In those, one block fills each of the two gaps between C and c, as in the diagram:



Ask yourself Why is this simpler than it looks?

With C and c opposite, the only freedom left is which block sits on which side (2 ways) and the order within each block ($2 \times 2 = 4$ ways). That gives $2 \times 4 = 8$ arrangements. Taking all three choices of the separated couple,

$$\mathrm{P}(\text{exactly two couples together}) = \frac{3 \times 8}{120} = \frac{24}{120} = \frac{1}{5}.$$

Part (iii)

Let E_k be the event that couple k sits together, for $k = 1, 2, 3$. We count each type of arrangement out of 120.

- $|E_1|$: glue couple 1 into a block. That leaves 5 units round the table, so there are $4! = 24$ arrangements, each with 2 internal orders, giving 48.
- $|E_1 \cap E_2|$: two blocks plus two people make 4 units. There are $3! = 6$ arrangements, with $2^2 = 4$ internal orders, giving 24.
- $|E_1 \cap E_2 \cap E_3| = 16$, from part (i).

By symmetry the same counts hold for any choice of couples. By inclusion–exclusion, the number of arrangements with at least one couple together is

$$|E_1 \cup E_2 \cup E_3| = 3(48) - 3(24) + 16 = 144 - 72 + 16 = 88.$$

So $120 - 88 = 32$ arrangements have no couple together, and

$$\mathrm{P}(\text{no husband next to his wife}) = \frac{32}{120} = \frac{4}{15}.$$

Ask yourself Can part (ii) be read off the same counts?

Yes. Each $|E_i \cap E_j|$ counts the arrangements with exactly the pair i, j together, plus those with all three together. So the number with exactly two couples together is $3(24) - 3(16) = 24$, matching part (ii).

Checks

- The four cases must account for all 120 arrangements. The count with exactly one couple together is $3(48) - 2 \cdot 3(24) + 3(16) = 48$, and $32 + 48 + 24 + 16 = 120$.
- A computer enumeration of all 720 labelled seatings gives probabilities $\frac{4}{15}, \frac{2}{5}, \frac{1}{5}, \frac{2}{15}$ for $0, 1, 2, 3$ couples together. These match the answers above.
- Sanity check: a given wife has two neighbours chosen from 5 people, so her husband is next to her with probability $\frac{2}{5}$. This agrees with $|E_1|/120 = 48/120$.

Common slips

- Using 6! arrangements but counting the favourable seatings as circular arrangements, or the other way round. The numerator and denominator must use the same convention.
- Forgetting the factor of 2 for the order of the two people within each block.
- In part (ii), forgetting to exclude the arrangements where the third couple also sits together. Without that, $\frac{3 \times 24}{120} = \frac{3}{5}$ is far too big.

Beyond STEP

This is the *ménage problem* in miniature. The classical version also requires men and women to alternate, and its general answer (Touchard's formula) comes from exactly this kind of inclusion–exclusion. In our unrestricted version with n couples, the expected number of couples sitting together is $n \cdot \frac{2}{2n-1} \to 1$. The probability that no couple sits together tends to e^{-1}, just like the famous derangement limit. In the alternating ménage version it tends to e^{-2}. Inclusion–exclusion and such limits are treated well in Grimmett and Welsh, *Probability: An Introduction*.

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